

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): DHD410S

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.04.0001-b002	2.3.1

Version History

Module Version	Date	Notes
1_0_0_0	9/19/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='DHD410S')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='DHD410S')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Auto' '16:10'	Value '4:3'	Value '16:9'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command ColorMode	Value 'Bright' 'Game'	Value 'PC' 'User'	Value 'Movie'
# ColorMode example InterfaceName.Set('ColorMode', 'Bright')			
Command DigitalZoom	Value 'Up'	Value 'Down'	
# DigitalZoom example InterfaceName.Set('DigitalZoom', 'Up')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'VGA 1' 'HDMI 2'	Value 'VGA 2' 'Video'	Value 'HDMI 1' 'Multimedia'
# Input example InterfaceName.Set('Input', 'VGA 1')			
Command MenuNavigation	Value 'Up' 'Down'	Value 'Left' 'Menu'	Value 'Right'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command Mute	Value 'On'	Value 'Off'	
# Mute example InterfaceName.Set('Mute', 'On')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Volume	Value 0 to 30 in steps of 1		

```
# Volume example  
InterfaceName.Set('Volume', 30)
```

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
```

FeedbackHandler will be called when any qualifier gets a new status.

Command AspectRatio	Value 'Auto' '16:10'	Value '4:3'	Value '16:9'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command ColorMode	Value 'Bright' 'Game'	Value 'PC' 'User'	Value 'Movie'
# ColorMode example InterfaceName.Update('ColorMode') Value = InterfaceName.ReadStatus('ColorMode') InterfaceName.SubscribeStatus('ColorMode', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Input	Value 'VGA 1' 'HDMI 2' 'None'	Value 'VGA 2' 'Video'	Value 'HDMI 1' 'Multimedia'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampUsage	Value Hours		
# LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power')			

```
InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)
```

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

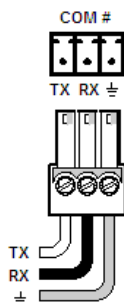
Data Bits: 8

Parity: None

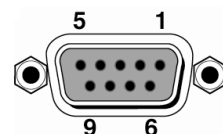
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



RS232 DB9, Female

Appendix A. Set Commands

Aspect Ratio 16:10	(ASP 4)\x0D
Aspect Ratio 16:9	(ASP 3)\x0D
Aspect Ratio 4:3	(ASP 2)\x0D
Aspect Ratio Auto	(ASP 1)\x0D
Auto Image None	(RSC 1)\x0D
AV Mute Off	(AVM 0)\x0D
AV Mute On	(AVM 1)\x0D
Color Mode Bright	(PST 1)\x0D
Color Mode Game	(PST 4)\x0D
Color Mode Movie	(PST 3)\x0D
Color Mode PC	(PST 2)\x0D
Color Mode User	(PST 5)\x0D
Digital Zoom Down	(DZM 0)\x0D
Digital Zoom Up	(DZM 1)\x0D
Freeze Off	(FRZ 0)\x0D
Freeze On	(FRZ 1)\x0D
Input HDMI 1	(SIN 3)\x0D
Input HDMI 2	(SIN 4)\x0D
Input Multimedia	(SIN 6)\x0D
Input VGA 1	(SIN 1)\x0D
Input VGA 2	(SIN 2)\x0D
Input Video	(SIN 5)\x0D
Menu Navigation Down	(KEY 4)\x0D
Menu Navigation Left	(KEY 2)\x0D
Menu Navigation Menu	(KEY 5)\x0D
Menu Navigation Right	(KEY 3)\x0D
Menu Navigation Up	(KEY 1)\x0D
Mute Off	(MUT 0)\x0D
Mute On	(MUT 1)\x0D
Power Off	(PWR 0)\x0D
Power On	(PWR 1)\x0D
Volume 0	(VOL 0)\x0D
Volume 30	(VOL 30)\x0D

Appendix B. Update Commands

Aspect Ratio	(ASP ?)\x0D
Power	(SST ?)\x0D

